Part 1: Text Processing and Exploratory Data Analysis

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GitHub Repository: https://github.com/martinafrisach/IRWA-2026-Final-Project
Submission TAG: [Actual tag name]

1. Overview

In Part 1, we pre-processed the fashion_products_dataset.json and performed exploratory data analysis (EDA) to prepare the data for indexing.

2. Dataset

We looked at the dataset, and we discovered that the dataset contains 28.080 products with 24 fields. And out this fields I pre-processed the following:

- pid
- title
- description
- brand
- out_of_stock
- selling_price
- discount
- actual_price
- average_rating

3. Text Pre-processing

This part takes a raw text which contains punctuation, inconsistent casing, and common words which do not contribute to search relevance. To solve this we implement a function build_terms(line) that applies five steps to each text string,

First, we convert all text to lowercase to ensure that words like "Shirt" and "shirt" are treated identically during search.

Second, we removed all punctuation marks which eliminates characters such as commas, periods, and exclamation marks.

Third, we tokenized the text by splitting it into individual words.

Fourth, we removed stop words, which are common words like "a", "the", and "is" that appear frequently but don’t carry any meaning for search purposes. And, also we thought of adding some extra stop words and we found the most relevant, but since the most appears only in the approximate 60% of the documents, we decided against it.

Finally, we applied stemming using the Porter Stemmer, which reduces words to their root form. For example, "cutting" and "cuts" all become "cut". This helps to group together words that have the same meaning, improving the recall of the search engine.

This preprocessing function was applied to both the title and description fields of each product, and we were inspired by the lab 1 we did in the subject.

4. Handling Structured Fields

For the fields category, sub_category, brand, product_details, and seller, we had to decide whether to merge them into a single text field, index them as separate fields, or handle them differently.

4.1. Option A: Merging into a Single Text Field

The first option is to merge all these fields into a single text field and index it as one. The main advantage of this approach is that it is easier to implement. The index structure will be simpler, and it won't be needed to manage multiple fields. However, this approach has significant cons. By merging distinct fields, we lose the ability to weight by specific field during ranking. For example, a match in the brand field might be more important than a match in the seller field, but merging prevents us from making this distinction.

4.2. Option B: Indexing as Separate Fields

The other option is to index each field separately in the inverted index. This option allows to preserve the distinctiveness of each field and allows that during the ranking it gives more precise results for each specific field . For example, a match in the brand field can be weighted in more detail than a match in the product details field. However, this option is more complex to implement and keep updated. The index becomes larger because each term is stored multiple times, once for each separate field. Moreover, ranking becomes more complex because we need to combine scores from multiple fields.

4.3. Option C: Hybrid Approach

A third option is a mix approach where similar fields are merged and others are kept separate. For example, category and sub_category could be merged because they are related, while brand and seller should be kept separate because they have a completely different meaning. This approach balances the cons of the first two options but needs a careful analysis to decide which fields to merge.

4.4. My Decision

After analyzing the data, we thought that the hybrid approach would be the most accurate. .

First, category and sub_category are closely related fields. The sub_category is essentially a more detailed version of the category. And since they are so similar they may contain overlapping vocabulary. Therefore, the best decision is merging them into a single field. This reduces the size of the index without losing relevant information.

Whereas brand, and seller refer to the opposite. The brand is the one that makes or designs the product, while the seller is the offering the product to the clients. Merging these fields would introduce noise because a search for a specific company would match products where this company appears only as a seller, even if the product itself is its product. Therefore, brand and seller should be kept as separate fields.

At last, product_details contains structured information such as color, fabric, and pattern. This field has a distinct vocabulary, so it should be kept separate because it contains information that is different from category, brand, or seller.

5. Handling Numeric Fields: Out-of-stock, Selling Price, Discount, Actual Price, and Average Rating

First, we decided that these fields don't need to be indexed as textual terms. Because these values are numeric or boolean (2 categories), and are better for using it as a numerical filter or ranking, than trying to find the match in the text. Indexing these fields as text would introduce noise and reduce retrieval effectiveness.

Textual indexing is designed for matching words, not numbers. By treating these fields as numeric, which may represent prices, ratings, percentages, we can implement them for efficient filtering, which improves the experience and the relevance of search results. For instance, products with higher ratings are ranked higher in search results.

To prepare these fields, we created a function to parse numeric values from string representations. For instance, the string "1,499" was converted to the number 1499.0.

After executing the function, we observed that almost every field was complete, with only little missing values. We didn't try to change these NaN values.

6. Exploratory Data Analysis (EDA)

6.1. Programming Methodology

First, we examined the structure of the dataset by checking for null values and duplicate pids, as well as calculating the distribution of products across categories, subcategories, brands, and sellers. For this task, we mainly used Pandas functions such as value_counts() and describe() to obtain summary statistics and determine how many products belong to each feature.

For the text analysis, we calculated the original word counts by splitting each text into words and measuring its length using len(). For the preprocessed text, we used len() on the lists of terms generated by the build_terms() function. To calculate vocabulary size, we collected the terms from all products and used sets to obtain the unique terms appearing in titles, descriptions, and both fields combined. We also used Counter to count how many times each processed term appeared across the dataset.

Finally, we analyzed stock availability using the out_of_stock column. We used value_counts() to count products marked as True or False and calculated their percentages.

6.2. Dataset Overview and Data Quality

The exploratory data analysis (EDA) makes us to better understand the dataset, identify data quality issues and analyze the characteristics of the dataset, in this case, the characteristics of the products.

The dataset contains 28.080 fashion products with 17 original attributes. After the preprocessing, 7 additional columns were created, resulting in 24 columns.

Although the null analysis gives us that the dataset does not contain any null value, we found that 11,150 products (that is 39.7%) have empty descriptions. This is an important fact because descriptions can provide useful information about the products for matching queries. Therefore, product titles and other attributes should also be considered during indexing.

We also checked for duplicate product identifiers (pid) and found no duplicates.

6.3. Product Distribution

We analyzed the distribution of products across categories, subcategories, brands and sellers to understand the composition of the dataset.

6.3.1 Category and Subcategory

The dataset is highly imbalanced. Clothing and Accessories represents 96.57% of the dataset, while Footwear is only 3.28%. The remaining categories represent very few products.

A similar pattern happens in subcategories, where the most common subcategory is Topwear (54.5%), followed by Bottomwear (13.1%) and Winter Wear (8.9%).

This imbalance can cause the search engine will mainly retrieve products related to clothes, while other product categories will have less representation.

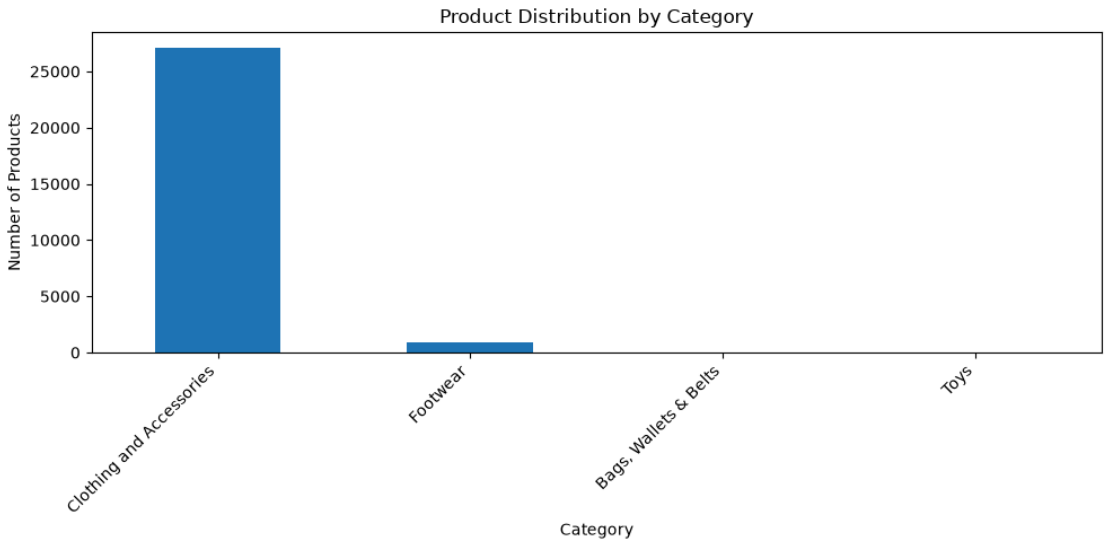


Figure 1: Product Distribution by Category

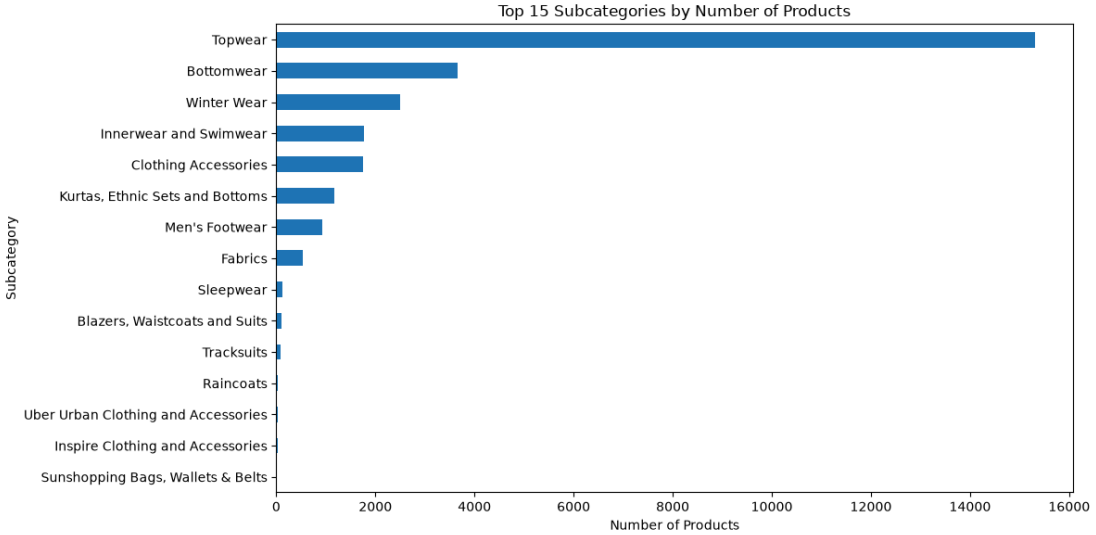


Figure 2: Top 15 Subcategories by Number of Products

6.3.2 Brand and Seller Distribution

We identified 325 brand values and 535 seller values, including empty labels.

A total of 2,009 products (7.2%) have no specified brand, while 1,643 products (5.85%) have no specified seller.

The distribution is uneven, as some brands and sellers appear much more frequently than others. For example, ECKO Unl, Free Authori, and ARBO are the most common brands, while RetailNet, SandSMarketing, and BioworldMerchandising are the sellers with the highest number of products.

This is relevant for the search engine because brands and sellers can help users find more specific products. For example, a user might search for a particular brand or want to filter results by seller. Therefore, keeping these attributes as separate fields would allow more precise filtering and ranking.

Finally, the missing brand and seller values should be handled carefully. We cannot exclude them from the search results. It is better to make these products with no information be searchable through their titles, descriptions or other characteristics.

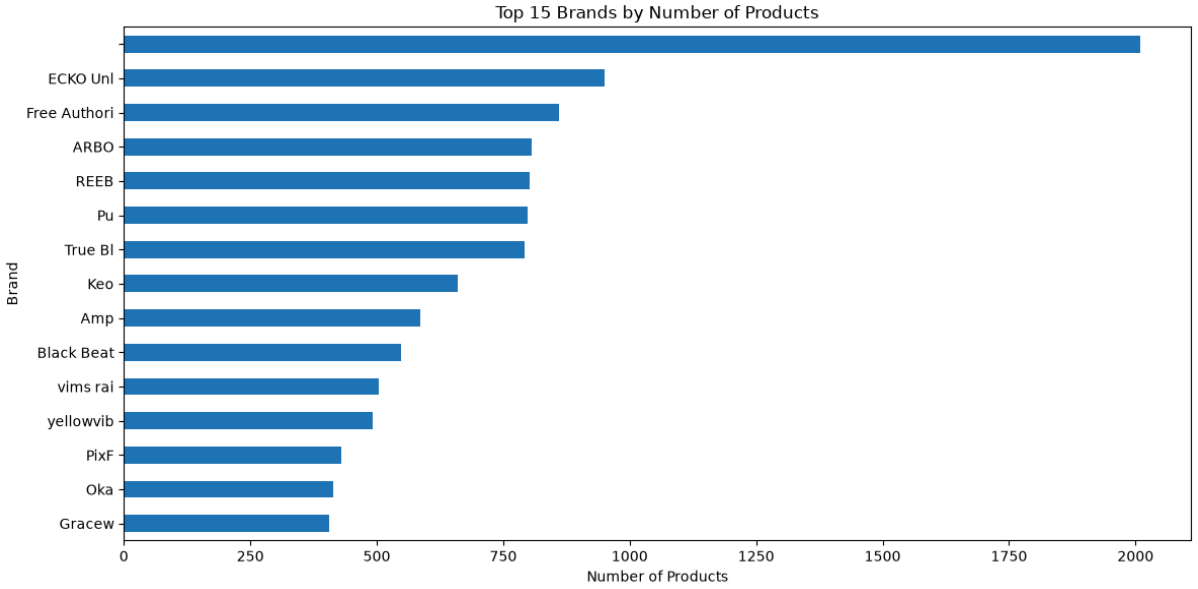


Figure 3: Top 15 Brands by Number of Products

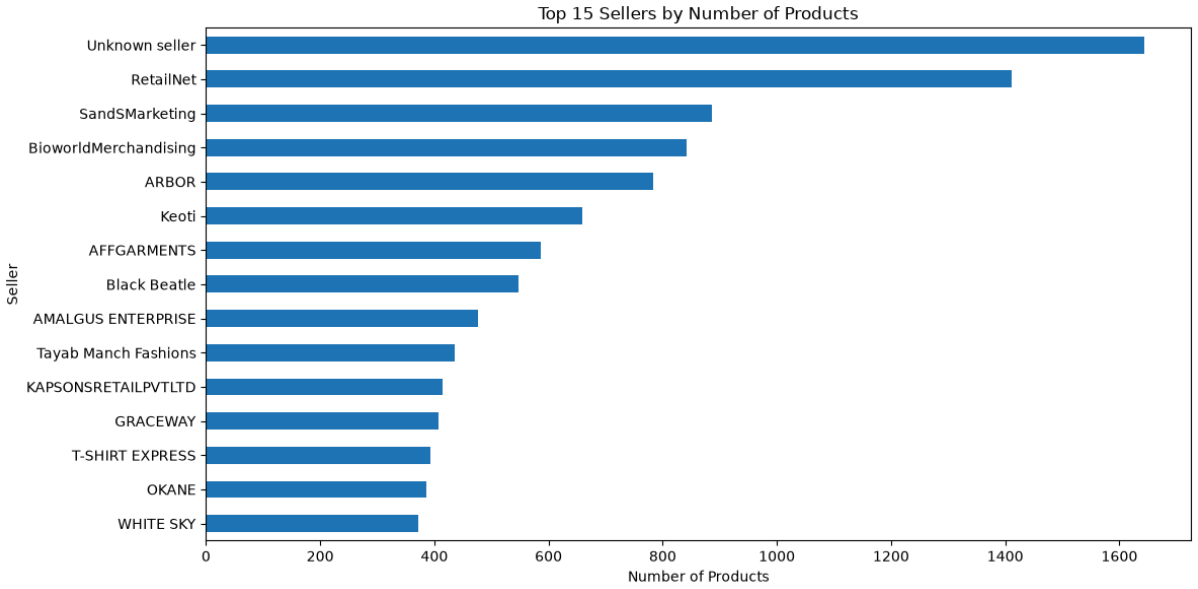


Figure 4: Top 15 Sellers by Number of Products

6.4. Text Analysis

This analysis was done to understand the characteristics of the product titles and descriptions and evaluate if they are useful for indexing.

6.4.1 Word Count Distribution

This analysis is interesting because for an information retrieval system, the length and quality of the text can influence how well products are represented in the search index.

Product titles contain an average of 6.46 words, while descriptions contain an average of 29.77 words.

After preprocessing, these averages decrease to 6.27 terms for titles and 18.84 terms for descriptions. This is due to the fact that we have removed stopwords and other non-informative elements. The decrease is much more noticeable in descriptions because they generally contain longer sentences and more common words.

The histograms reveal that most product titles have more or less the similar length, between 5 and 8 words. In contrast, description lengths are much more variable. Many products have shorter descriptions or with no descriptions while less products contain large descriptions. This produces a right-skewed distribution.

Additionally, 39.7% of products have no description terms. This is an important limitation because these products cannot be matched by description. However, their titles and other characteristics may still have enough information to make them searchable.

These results tell us that titles have a lot of importance in the search engine, as they are short, available and contain relevant information about the product. Descriptions can be a complement providing additional vocabulary when available.

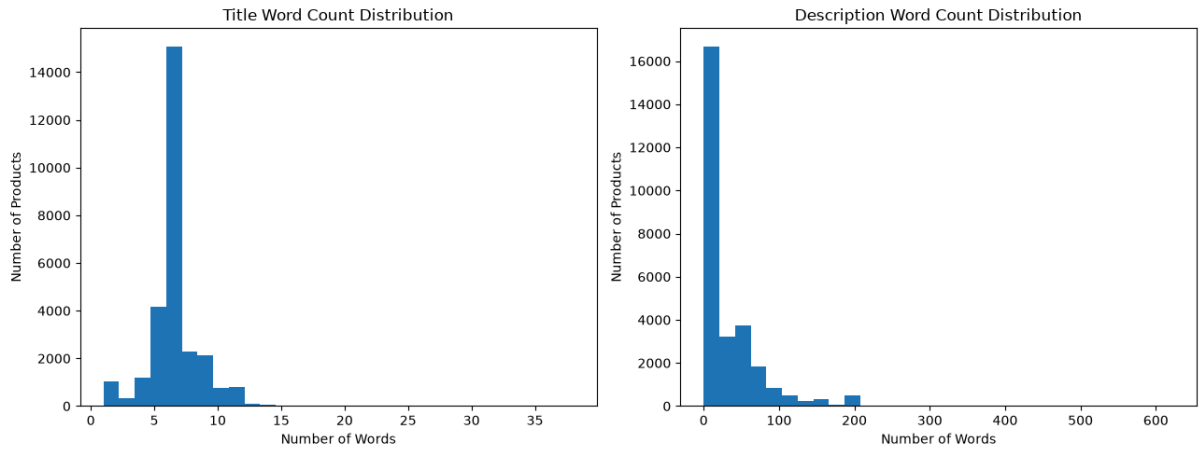


Figure 5: Title and Description Word Count Distribution

6.4.2 Vocabulary Size

After analyzing text length, we calculated the vocabulary size of product titles and descriptions after preprocessing.

Vocabulary size refers to the number of unique terms appearing in a collection of texts. This is important for an inverted index because the index needs to store the different terms that users may search for.

The results were:

- Title vocabulary: 738 unique terms.
- Description vocabulary: 4,698 unique terms.
- Combined vocabulary: 4,902 unique terms.

These results show that descriptions contain the most variety of terms than titles. This makes sense since titles are more concise and follow similar patterns while descriptions contain additional information about other characteristics.

These results show that descriptions provide valuable vocabulary diversity, even though a significant number of products have empty descriptions. Therefore, using both titles and descriptions is preferable to relying on only one of these fields.

6.4.3 Most frequent terms

We also analyzed the most frequent terms in product descriptions after preprocessing.

We wanted to identify which words appear most often and see the main characteristics represented in the dataset.
The most frequent terms include:

- shirt: 18,321 occurrences.
- cotton: 10,139 occurrences.
- wear: 9,353 occurrences.
- comfort: 8,836 occurrences.

These results make sense since we have seen before that the most common category and subcategories were clothing and topwear.

We also observed that some terms appear in shortened forms, such as sleev or qualiti. This happens because the Porter Stemmer reduces words to their stems during preprocessing. However, they are relevant for matching related word forms during retrieval.

An important consideration is that very frequent terms are not always the most useful for distinguishing between products.

For example, if the word shirt appears in a large number of documents, it provides less information for make differences between products than a more specific term.

Overall, the most frequent terms confirm that the textual content is closely related to the products and provide useful information into the vocabulary that will be indexed.

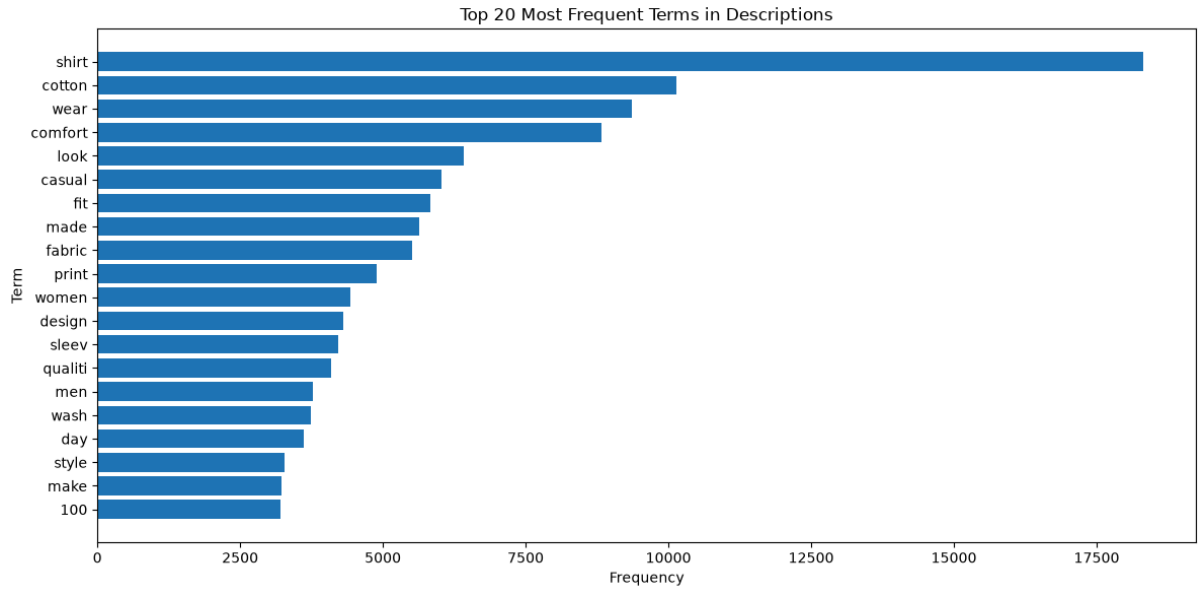


Figure 6: Top 20 Most Frequent Terms in Descriptions

6.5. Price, Discount and Rating Analysis

The dataset also contain numerical columns that can be useful for filtering, sorting and ranking products.

We analyzed selling price, discount and average customer rating.

These attributes can provide additional information to improve the user experience in the search engine.

6.5.1 Price distribution

Analyzing the selling price helps us to understand the price range of the products.

The average selling price is 705.64, while the median is 545. The minimum recorded selling price is 99, and the maximum is 7,999.

With these values and the distribution plot we can assess that the distribution is right-skewed. This means that most of the products have low por moderate prices while a few of them increase the average price.

Additionally, we identified 2 products with missing selling-price values.

Since price is a common criteria for looking for products, this is really useful. For example, a user might want to find products below a certain price, search within a specific budget, or sort the results from cheapest to most expensive.



Figure 7: Distribution of Product Selling Prices

6.5.2 Discount distribution

The average discount is 50.26%, while the median is 53%. Additionally, 50% of the products with recorded discounts have values between 40% and 63%. The maximum recorded discount is 87%, while the minimum is 1%.

The histogram shows that discounts are mainly concentrated around 40% to 65%, with fewer products having very low or extremely high discounts. This shows us that high discounts are common in the dataset.

We also identified 855 products without a recorded discount value.

This is relevant for the search engine because discounts could be used as an additional filtering or sorting criterion.

As we know, discount is highly related with price, so for this reason, discount information should be considered together with selling price and other attributes.

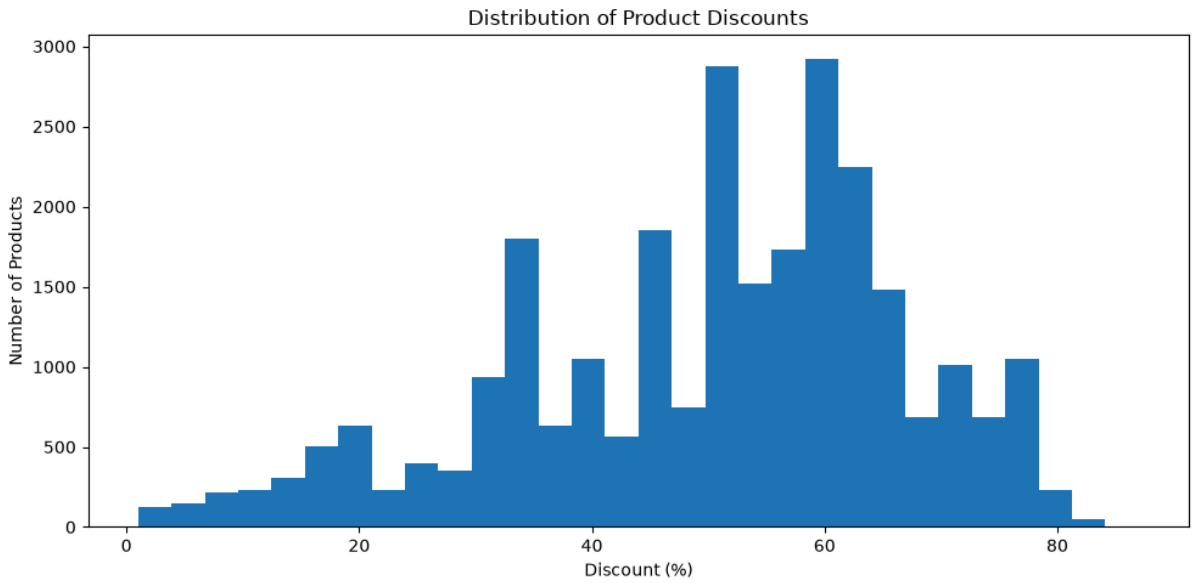


Figure 8: Distribution of Product Discounts

6.5.3 Rating distribution

Average ratings of users helps us to understand how products are evaluated and if they can provide useful information for the search engine.

The average rating is 3.63, while the median is 3.8. Most ratings are concentrated between approximately 3 and 4.5, suggesting that many products receive moderately positive evaluations.

We also found that 2,261 products have no recorded rating.

As the other features, this is useful for filtering or sorting criteria. Since we don't know how many users contributed to each ranking, a high rating is not necessarily more reliable than another product.

Therefore, ratings can provide useful complementary information, but they should not be considered a complete measure of product quality.

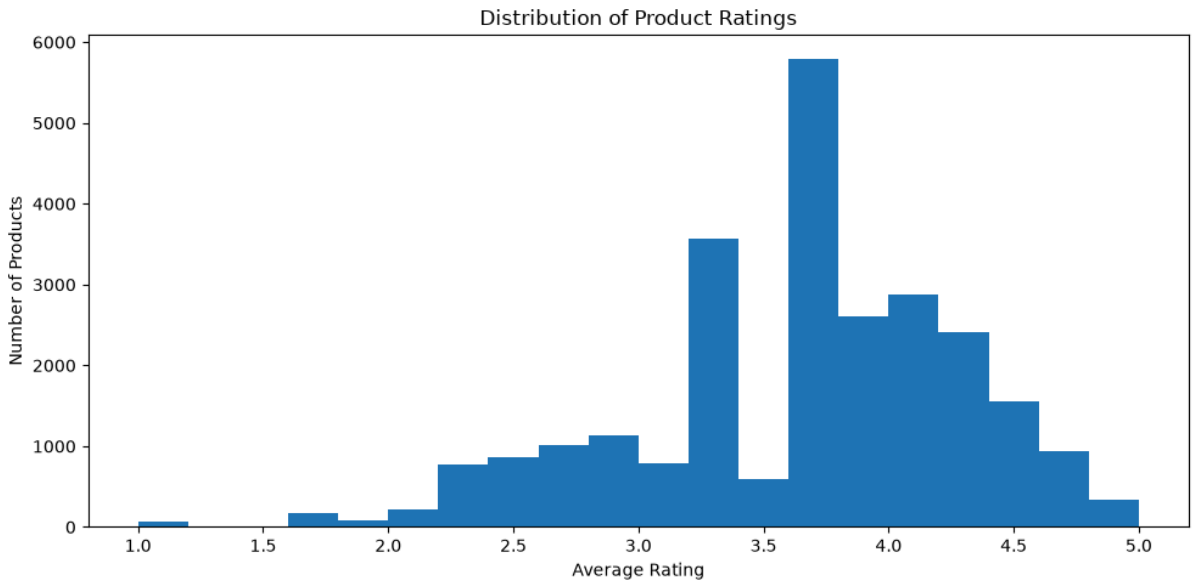


Figure 9: Distribution of Product Ratings

6.6. Product Rankings

After this numerical analysis, we made 3 product rankings to explore the highest values of different characteristics. They allow us to examine individual products rather than only general statistics. It also helps to illustrate possible sorting options that could be included in the search engine.

6.6.1 Top Rated Products

All products in this ranking have a rating of 5.0, which is the maximum possible value. However, their selling prices vary considerably. This shows that products with the same rating can belong to different price ranges. However, as we mentioned before, we don't know the number of customers that have rated.

For example, a product with a rating of 5.0 based on a single review may be less reliable than a product with a slightly lower rating based on many reviews.

Therefore, this ranking should be interpreted as a list of products with the highest recorded average ratings, rather than a definitive ranking of the best products.

	title	brand	average_rating	selling_price
27767	Solid Women Round Neck Blue T-Shirt	Oka	5.0	385.0
12332	Printed Women Hooded Neck Black T-Shirt	ATTIITU	5.0	549.0
12279	Printed Women Hooded Neck Grey T-Shirt	ATTIITU	5.0	909.0
23852	Graphic Print Men Round Neck Blue T-Shirt	Free Authori	5.0	519.0
12235	Solid Women Round Neck White, Black T-Shirt	ATTIITU	5.0	649.0
12242	ATTIITUDE Men Vest		5.0	937.0
12243	ATTIITUDE Men Vest		5.0	937.0
8477	Women Trunks	V	5.0	420.0
1923	Self Design Women Black Track Pants	REEB	5.0	2659.0
18049	Printed Women Round Neck White T-Shirt	yellowvib	5.0	870.0

Figure 10: Top Rated Products Table

6.6.2 Most Expensive Products

Their prices range from 6,493 to 7,999, placing them above the median selling price of 545. This confirms the existence of a small group of expensive products, which explains the right-skewed price distribution observed before.

We also compared their customer ratings and found that they vary considerably. This suggests that a higher selling price does not necessarily correspond to a higher customer rating. Price and rating represent different characteristics of a product and should therefore be treated as separate attributes.

	title	brand	selling_price	average_rating
2067	Solid Women Track Suit	REEB	7999.0	2.7
11010	Checkered Single Breasted Party Women Full Sle...	True Bl	7998.0	2.6
25815	Full Sleeve Solid Women Casual Jacket	Pu	7799.0	3.7
26089	Full Sleeve Printed Men Sweatshirt	Pu	7799.0	4.5
6895	Skinny Men Blue Jeans	G	7794.0	4.2
11008	Self Design Single Breasted Party Women Full S...	True Bl	6998.0	2.6
6898	Regular Fit Men Dark Blue Cotton Blend Trousers	G	6925.0	4.5
6901	Maxx Regular Men Black Jeans	G	6692.0	3.8
25423	Full Sleeve Solid Women Casual Jacket	Pu	6499.0	3.7
6855	Skinny Men Blue Jeans	G	6493.0	NaN

Figure 11: Most Expensive Products Table

6.6.3 Products with the Highest Discounts

The discounts in this ranking range from 84% to 87%, which are considerably higher than the dataset's average discount of 50.26%. Most of these products are in the T-shirt category, which is

consistent with what we saw before. We also observed that brands such as Jack Roy and yellowvib appear among the products with the highest discounts.

	title	brand	discount	selling_price
906	Printed Women Round Neck White T-Shirt	Jack Roy	87.0	128.0
902	Printed Women Round Neck Grey T-Shirt	Jack Roy	86.0	136.0
903	Printed Women Round Neck White T-Shirt	Jack Roy	86.0	136.0
18249	Printed Men Round Neck Multicolor T-Shirt (Pa...	yellowvib	85.0	449.0
9813	Solid Balclava Cap	Gracew	84.0	228.0
91	nu-Lite Satin Tie & Cufflink (Red)		84.0	499.0
18093	Printed Men Collared Neck Multicolor T-Shirt	yellowvib	84.0	230.0
520	Printed Women Round or Crew Black, Grey T-Shirt	Fairdea	84.0	201.0
18016	Printed Women Round Neck Blue T-Shirt	yellowvib	84.0	280.0
18017	Printed Men Mandarin Collar Blue T-Shirt	yellowvib	84.0	282.0

Figure 12: Products with the Highest Discounts Table

6.7. Stock Availability

Finally, we analyzed the out_of_stock field to understand how many products are currently available for purchase.

We can see that 26,436 products (94.15%) are in stock, while 1,644 products (5.85%) are out of stock. So the majority of products in the dataset are available. This field is really important for a search engine since the user may want to be bought.

Stock availability should be a filtering requirement or should be used to prioritize available products in the search results.

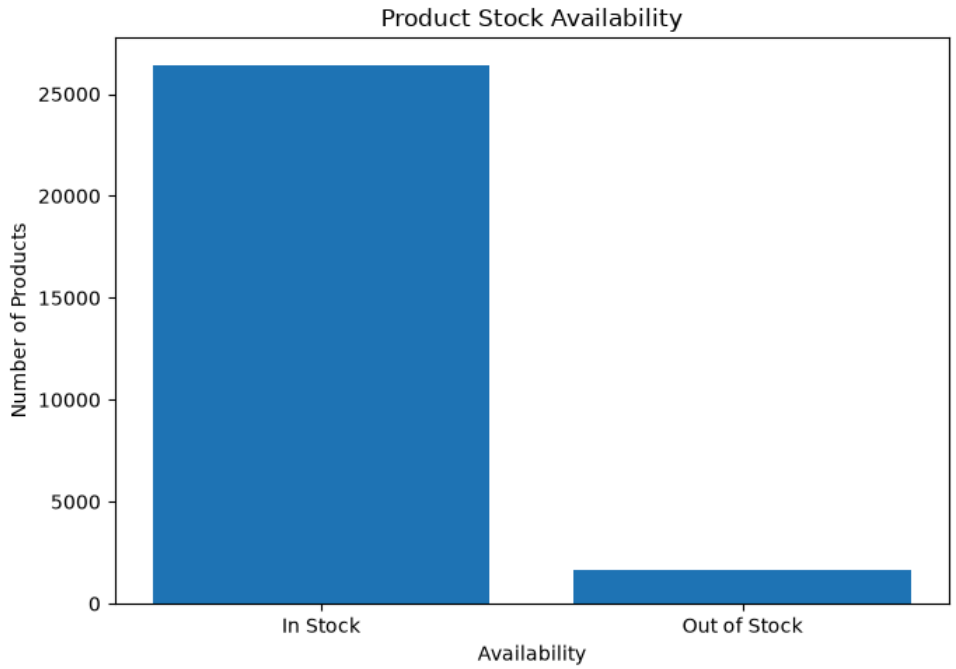


Figure 13: Product Stock Availability

7. Main Conclusions

In this first part of the project, we performed text preprocessing and exploratory data analysis on a dataset containing 28,080 fashion products to prepare the information for the search engine.

The steps that we made like lowercasing titles and descriptions or tokenization help normalize the text and make it easier to match products with user queries.

The EDA showed that the dataset is highly imbalanced, with most products belonging to Clothing and Accessories, especially Topwear. We also found that 39.7% of products have empty descriptions, which highlights the importance of using titles and other attributes during retrieval. However, descriptions provide a richer vocabulary than titles, although titles are shorter. Therefore, combining both fields can improve product representation.

Finally, we observed that prices have a right-skewed distribution, discounts are generally high, and most products have moderately positive ratings. We also found that 94.15% of products are in stock. These attributes could be useful for improving search results through filtering and sorting.

Overall, this first part of the project helped us understand the dataset, identify its limitations, and make important decisions for helping us to do and think about the indexing.

AI USAGE
We used AI to review the report draft and debug some code.